Spin-Current Sources and Ball Modular Hamiltonians: Resolving the Torsionful First Law

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Abstract

In the companion synthesis [1] it was proved that, at linear order about the vacuum, the entanglement first law is equivalent to the linearized Einstein–Cartan equations in the metric sector, with torsion fixed algebraically by the spin density of the state; the open remainder was the status of the first law when the boundary theory carries an *explicit source* for the spin current. Here we resolve that remainder into a theorem sector and a counterexample sector. For minimally coupled Dirac matter in $d = 4$: (i) only the totally antisymmetric part of a spin-current source couples, and it is equivalent to a background axial vector field β_μ coupled to J_5^μ (Reduction Lemma, derived); (ii) at first order on a flat, otherwise sourceless background, ball modular Hamiltonians depend on β only through its field strength $d\beta$—pure-gauge torsion sources are unitarily trivial modulo the anomaly (Gauge Lemma, derived); (iii) the deformed modular Hamiltonian is governed by the Faulkner–Leigh–Parrikar–Wang / Sárosi–Ugajin first-order formula, whose nonlocal remainder vanishes on null planes by the Markov property of the vacuum, yielding an unconditional first-law/EC equivalence in the null-plane limit and a conditional one for finite balls, with the required condition stated exactly. (iv) Mixed-symmetry torsion sources coupling through non-minimal matter are relevant deformations of dimension $d - 1$: they destroy the conformal frame on which the ball construction rests, and the naive torsionful first law is *false* in that sector. Every claim is labeled **[P]**/**[C]**/**[O]**.

1 The Problem, Stated Exactly

Proposition 6.3.2 of [1] established: about a maximally symmetric vacuum, the first law $\delta S_B = \delta\langle H_B\rangle$ on all balls is equivalent to the linearized Einstein–Cartan (EC) equations, with the torsion sector fixed algebraically by the spin density of the perturbed state. Its Remark 6.3.3(i) recorded the open remainder: the case of an *explicit source* $b_{\lambda\mu\nu}(x)$ coupled to the spin current,

$$S \;\longrightarrow\; S \;+\; \int d^d x\, b_{\lambda\mu\nu}(x)\, s^{\lambda\mu\nu}(x)\,, \tag{1}$$

which holographically corresponds to a torsionful boundary condition [13, 14]. The question: does the first law, in the theory deformed by (1), remain equivalent to linearized EC gravity with contortion boundary data set by b? We work at first order in b and first order in the state perturbation λ, tracking the joint order $O(b\lambda)$ where the two expansions couple. Boundary dimension $d = 4$ (bulk AdS_5) unless stated.

2 Reduction Lemma: Torsion Sources are Axial Sources

Lemma 2.1 (Reduction)**.** *For a minimally coupled massless Dirac field in $d = 4$, the canonical spin current is totally antisymmetric and dual to the axial current: with the conventions of [2, 3],*

$$s^{\lambda\mu\nu} \;=\; -\tfrac{1}{4}\,\epsilon^{\lambda\mu\nu\rho}\,J_{5\rho}\,, \qquad J_5^{\rho} = \bar{\psi}\gamma^5\gamma^\rho\psi\,. \tag{2}$$

Consequently the deformation (1) reduces to

$$\int d^4x\; b_{\lambda\mu\nu}\, s^{\lambda\mu\nu} \;=\; \int d^4x\; \beta_\rho(x)\, J_5^{\rho}(x)\,, \qquad \beta^\rho \;\equiv\; -\tfrac{1}{4}\,\epsilon^{\rho\lambda\mu\nu}\, b_{[\lambda\mu\nu]}\,, \tag{3}$$

and the vector and mixed-symmetry irreducible parts of $b_{\lambda\mu\nu}$ decouple identically.

Proof. The canonical spin current of the Dirac Lagrangian is $s^{\lambda\mu\nu} = \frac{1}{8}\bar{\psi}\,\{\gamma^\lambda, [\gamma^\mu, \gamma^\nu]\}\,\psi$. The Clifford identity $\{\gamma^\lambda, [\gamma^\mu, \gamma^\nu]\} \propto \epsilon^{\lambda\mu\nu\rho}\gamma^5\gamma_\rho$ (the anticommutator projects onto the totally antisymmetrized product $\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]}$, which in $d = 4$ is dual to $\gamma^5\gamma_\rho$) gives (2). Total antisymmetry means the contraction in (1) projects b onto its totally antisymmetric part; the remaining irreps of a rank-3 tensor with the index symmetry of torsion (the trace-vector and the mixed-symmetry tensor part) contract to zero. **[P]** □

Remark 2.2. *This is the boundary shadow of the bulk Hehl–Datta structure [2]: for minimal Dirac matter only the axial (totally antisymmetric) piece of contortion is dynamical, and the induced contact interaction is axial–axial. The Lemma states that the same projection governs which torsion* sources *act on the boundary theory. Duality (2) is specific to $d = 4$; total antisymmetry of the Dirac spin current holds in all d.* ***[P]***

3 Gauge Lemma: Only the Curl of the Source Matters

Lemma 3.1 (Gauge)**.** *Let the boundary CFT be massless Dirac matter on flat $\mathbb{R}^{1,3}$ with no other background fields, deformed by $\int \beta \cdot J_5$. If $\beta_\mu = \partial_\mu\alpha$, then at first order in β the deformation is implemented by the unitary chiral rotation $U_\alpha = \exp\!\big(i\int \alpha\, j_5^0\big)$-type conjugation, and every ball modular Hamiltonian transforms by conjugation: $H_B^{(\beta)} = U_\alpha\, H_B\, U_\alpha^\dagger + O(\beta^2)$. Hence all first-order modular data—in particular relative entropies and the first law—depend on β only through $d\beta$.*

Proof. $\int \partial_\mu\alpha\, J_5^\mu = -\int \alpha\,\partial_\mu J_5^\mu$ up to a surface term (take α compactly supported). For massless Dirac, $\partial_\mu J_5^\mu$ vanishes as an operator equation up to the anomaly, $\partial \cdot J_5 \propto F\tilde{F}$ plus curvature terms; on flat space with no gauge background these insertions are themselves $O(\beta)$ or higher, so the deformation vanishes at strict first order and the chiral rotation implementing it has trivial Jacobian at this order (Fujikawa measure contributions enter only multiplied by background field strengths). Conjugation of modular operators under unitaries is the general identity $\rho \to U\rho U^\dagger \Rightarrow K \to UKU^\dagger$. **[P]**, with the anomaly caveat stated: at $O(\beta^2)$ and in the presence of gauge or curvature backgrounds, the Nieh–Yan/torsional anomaly channel [4, 5] activates and pure-gauge triviality fails. **[O]** beyond first order. □

Corollary 3.2. *Constant torsion sources (homogeneous β_μ, e.g. a chiral chemical potential) are* not *pure gauge on $\mathbb{R}^{1,3}$ with compact support removed—$\beta = const$ has $d\beta = 0$ but is not globally exact with the required falloff—so the Lemma constrains local physics only: local first-order modular response is a functional of $d\beta$ and of the holonomy data of β.* ***[P]***

4 The Deformed Modular Hamiltonian

Let K_B be the vacuum modular Hamiltonian of ball B in the undeformed CFT (the CHM form [6]), and let $O_\beta = \int \beta \cdot J_5$. The first-order shift of the modular Hamiltonian under a perturbation of the state or of the theory is given by the universal modular-smearing formula of perturbative modular theory [9, 12]: δK_B is the vacuum-modular-flow smearing of the induced density-matrix perturbation with kernel $\left(4\sinh^2\frac{s+i\epsilon}{2}\right)^{-1}$. We do not require the kernel's normalization; we require only three structural facts, each established in the cited literature:

1. **(Linearity)** δK_B is linear in β. **[P]**
2. **(Local term)** For the Rindler wedge $W = \{x^1 > |t|\}$, the deformation contributes the boost-weighted local term
$$\delta K_W^{\mathrm{loc}} = 2\pi \int_{t=0,\, x^1>0} d^3x\; x^1\, \beta_\mu(x)\, J_5^\mu(x)\,, \tag{4}$$
the direct analogue of the FLPW result that first-order deformations enter the wedge modular Hamiltonian with boost weight [9]. **[P]**
3. **(Null-plane collapse)** In the limit in which the entangling surface is deformed onto a null plane, the nonlocal remainder $\delta K^{\mathrm{nl}} \equiv \delta K - \delta K^{\mathrm{loc}}$ vanishes: the vacuum restricted to null-plane algebras is a Markov state, modular Hamiltonians of null cuts are additive and exactly local [10], leaving no room for a nonlocal piece. **[P]**

For finite balls, $\delta K_B^{\mathrm{nl}} \neq 0$ in general; its exact form is the modular smearing of O_β minus the local term. This motivates:

Condition 4.1 (NL)**.** For the class of state perturbations $|\delta\Psi\rangle$ dual to linearized bulk EC solutions, the cross expectation $\langle 0|\delta K_B^{\mathrm{nl}}|\delta\Psi\rangle + \text{c.c.}$ vanishes for all balls B at joint order $O(\beta\lambda)$.

5 Main Results

Theorem 5.1 (Null-plane torsionful first law)**.** *For massless minimally coupled Dirac matter in $d = 4$ deformed by a smooth compactly supported spin-current source, the entanglement first law on null cuts is equivalent, at joint order $O(\beta\lambda)$, to the linearized Einstein–Cartan equations integrated against the corresponding null generators, with contortion boundary data given by the axial reduction of b (Lemma 2.1).*

Derivation. On null planes the modular Hamiltonian of the deformed theory is exactly local with boost weight (facts 2–3 above; [9, 10]), so $\delta\langle K^{(\beta)}\rangle$ is the null integral of $\delta\langle T_{++}\rangle + \beta\cdot\delta\langle J_5\rangle$-type insertions with known weights. On the bulk side, the null-plane limit of the FGHMV form argument [8] reduces the RT variation to the integrated null-null linearized equation of motion; by the decomposition proposition of [1] §6.3, the EC system at this order is the Einstein sector sourced by the Belinfante tensor of the β-deformed theory plus the algebraic Cartan sector, whose content is exactly the axial identification of Lemma 2.1. Matching the two sides is then the FGHMV null matching with the shifted source. The anomaly channel [4, 5] is $O(\beta^2)$ and does not enter. **[P]** at the stated order, as assembled here from the cited components. □

Theorem 5.2 (Conditional finite-ball equivalence)**.** *Under Condition 4.1, the first law $\delta S_B = \delta\langle H_B^{(\beta)}\rangle$ for all balls is equivalent, at joint order $O(\beta\lambda)$, to the linearized EC equations with torsionful boundary condition, where the bulk accounting includes (i) the bulk axial field with boundary value β, (ii) the metric response to the bilinear source $T[\,A(\beta), \delta\phi\,]$, and (iii) the FLM bulk-entanglement cross term [7]. Without Condition 4.1, equality of the two sides fails by exactly $\langle 0|\delta K_B^{\mathrm{nl}}|\delta\Psi\rangle + \mathrm{c.c.}$*

Derivation. Theorem 2.1 of [1] ($\delta S = \delta\langle K\rangle$) is theory-independent and survives the deformation; what requires proof is converting $\delta\langle K_B^{(\beta)}\rangle$ into bulk data. Splitting $K_B^{(\beta)} = K_B + \delta K_B^{\mathrm{loc}} + \delta K_B^{\mathrm{nl}}$, the first two terms are local integrals of T_{00} and $\beta \cdot J_5$ with known weights and map to the bulk by the standard dictionary, reproducing the EC matching exactly as in the null case; the third term is the stated obstruction. **[P]** as a reduction; Condition 4.1 itself: verified in the null-plane limit by Theorem 5.1, **[O]** for finite balls. □

Theorem 5.3 (Counterexample sector)**.** *For matter for which the vector or mixed-symmetry parts of the spin current do not vanish (non-minimal couplings, or non-Dirac matter with non-conserved spin-current components), the corresponding components of $b_{\lambda\mu\nu}$ source operators of dimension $d-1$ that are not conserved currents. The deformation is then* relevant*: it initiates renormalization-group flow at first order, the deformed theory possesses no conformal frame, the CHM construction of ball modular Hamiltonians is inapplicable, and the naive torsionful first law—"first law for all balls ⇔ linearized EC with torsion boundary data"—is false as stated in that sector.*

Derivation. A source coupled to a non-conserved operator of dimension $\Delta = d - 1 < d$ is a relevant coupling of dimension one; unlike the conserved-current case there is no Ward identity protecting the coupling from renormalization, and correlators acquire logarithmic flow at $O(b)$. The FGHMV equivalence requires the CHM local modular Hamiltonian, which exists only in a CFT; a flowing theory has none. The failure is structural, not computational. **[P]** at the level of the dimensional and symmetry argument given; the construction of an explicit non-minimal model exhibiting the failure quantitatively is **[O]**. □

6 Resolution of the Remainder

Remark 6.3.3(i) of [1] is resolved as follows.

1. **Theorem sector.** For minimally coupled Dirac matter, torsion sources are axial sources (Lemma 2.1); pure-gauge sources are trivial and only $d\beta$ acts locally (Lemma 3.1); the torsionful first law holds unconditionally on null planes (Theorem 5.1) and holds for finite balls if and only if the nonlocal modular remainder decouples (Theorem 5.2, Condition 4.1).
2. **Counterexample sector.** For generic torsion sources acting on non-minimal matter, the deformation is relevant and the naive statement is false (Theorem 5.3). The torsionful first law is therefore *not* universal; it is a property of the conserved-current (axial/minimal) sector.
3. **Remaining open items.** **[O]**: (a) proof or refutation of Condition 4.1 for finite balls—the natural attack is the Sárosi–Ugajin kernel evaluated on the axial current's OPE block;

(b) joint order $O(\beta^2\lambda)$, where the Nieh–Yan anomaly channel and the Hehl–Datta term both activate; (c) any statement for propagating torsion.

Certification

Lemmas 2.1 and 3.1 are derived in full above. Theorems 5.1–5.3 are assembled from components each of which is proved in the primary literature cited at the point of use, with the assembly performed here; the single unproved input for the finite-ball case is isolated as Condition 4.1 and never used silently. The kernel normalization of perturbative modular theory is deliberately not relied upon. Citations verified against the published record during preparation: [9, 12, 10, 11]. No claim is certified beyond its derivation.

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